Gergely Buza

ETH Zürich
Rämistrasse 101, 8092 Zürich
Switzerland

buzag@ethz.ch
Personal webpage

Education

DAMTP, University of Cambridge October 2020 – September 2024
Ph.D. in Mathematics *Churchill College*
Supervisor: Rich Kerswell; Co-advisor: Edriss Titi
Thesis: **"A dynamical systems approach to the study of local phenomena in fluid flows"**

ETH Zürich September 2018 – June 2020
M.Sc. in Mechanical Engineering *Grade: 5.94/6.00*[1]
Advisors: Shobhit Jain & George Haller
Thesis: **"Model reduction for steady-state periodic response"**

Budapest University of Technology and Economics September 2014 – January 2018
B.Sc. in Mechanical Engineering *Grade: 4.95/5.00*[2]
Advisor: Tamás Insperger
Thesis: **"Balancing on a see-saw with reaction time delay"**

SZTE Ságvári Endre Gyakorló Gimnázium September 2010 – June 2014
Secondary School, Mathematics specialization

Temporary Positions

ETH Zürich July 2025 – Present
Postdoctoral Research Associate

DAMTP, University of Cambridge October 2024 – April 2025
Postdoctoral Research Associate

MTA-BME Lendület Human Balancing Research Group February 2018 – August 2018
Research Assistant

Research Interests

Dynamical systems, parabolic/hyperbolic PDEs, fluid dynamics

Publications

13. Buza, G. and Haller, G., 2026. *An application of the parameterization method to evolution flows associated to nonautonomous systems*. ArXiv preprint. arXiv:2610.18078.

12. Buza, G. and Haller, G., 2026. *Spectral submanifold reduction for PDEs describing nonlinear continuum vibrations*. ArXiv preprint. arXiv:2607.10675.

11. Abbasciano, G., Buza, G. and Haller, G., 2026. *Data-driven reduced modeling of delayed dynamical systems via spectral submanifolds*. Chaos: An Interdisciplinary Journal of Nonlinear Science 36(10). DOI: 10.1063/5.0344712.

[1] Switzerland uses a 6-point grading system, where 6 represents excellent, 4 pass and 1 very poor.
[2] Hungary uses a 5-point grading system, where 5 represents excellent, 2 pass and 1 very poor.

10. King, J., Kaszás, B., Buza, G., Jussiau, W. and Haller, G., 2026. *Parametric spectral submanifolds across Hopf bifurcations with applications to fluid dynamics*. Chaos: An Interdisciplinary Journal of Nonlinear Science 36(10). DOI: 10.1063/5.0334774.

9. Buza, G. and Haller, G., 2026. *Existence of spectral submanifolds in time delay systems*. Journal of Nonlinear Science, 36(5), p.109. DOI: 10.1007/s00332-026-10327-y.

8. Buza, G., 2025. *Smooth invariant foliations and Koopman eigenfunctions about stable equilibria of semiflows*. Journal of Differential Equations 440: 113447. DOI: j.jde.2025.113447.

7. Buza, G., 2024. *Spectral submanifolds of the Navier–Stokes equations*. SIAM Journal on Applied Dynamical Systems, 23(2), pp.1052-1089. DOI: 10.1137/23M154858X.

6. Buza, G., Beneitez, M., Page, J. and Kerswell, R.R., 2022. *Finite-amplitude elastic waves in viscoelastic channel flow from large to zero Reynolds number*. Journal of Fluid Mechanics, 951, p.A3. DOI: 10.1017/jfm.2022.831.

5. Buza, G., Page, J. and Kerswell, R.R., 2022. *Weakly nonlinear analysis of the viscoelastic instability in channel flow for finite and vanishing Reynolds numbers*. Journal of Fluid Mechanics, 940, p.A11. DOI: 10.1017/jfm.2022.222.

4. Buza, G., Haller, G. and Jain, S., 2021. *Integral equations and model reduction for fast computation of nonlinear periodic response*. International Journal for Numerical Methods in Engineering, 122(17), pp.4637-4659. DOI: 10.1002/nme.6740.

3. Buza, G., Jain, S. and Haller, G., 2021. *Using spectral submanifolds for optimal mode selection in nonlinear model reduction*. Proceedings of the Royal Society A, 477(2246), p.20200725. DOI: 10.1098/rspa.2020.0725.

2. Buza, G, Milton, J, Bencsik, L, Insperger, T, 2020. *Establishing metrics and control laws for the learning process: Ball and beam balancing*, Biological Cybernetics. DOI: 10.1007/s00422-020-00815-z.

1. Buza, G, Insperger, T, 2018. *Mathematical models for balancing tasks on a see-saw with reaction time delay*. 14th IFAC Workshop on Time Delay Systems, Budapest, Hungary, June 28-30, 2018. IFAC-PapersOnLine, 51(14):288-293, DOI: 10.1016/j.ifacol.2018.07.238.

Talks

NODYCON2026, Rome	September 2026
University of Szeged, Bolyai Institute, Differential Equations Seminar	May 2026
ETH Zürich, Nonlinear Dynamics Seminar	March 2026
University of Cambridge, DAMTP, Fluid Mechanics Seminar	February 2025
University of Bristol, Fluids and Materials Seminar	February 2025

Teaching Experience

DAMTP, University of Cambridge

Dynamical systems supervisions — *2021 – 2024*

Vector calculus supervisions — *2025*

ETH Zürich

Nonlinear Dynamics and Chaos I, jointly with George Haller — *2026*

Nonlinear Dynamics and Chaos II — *2027*

Awards & Honors

Harding Distinguished Postgraduate Scholarship — 2020 – 2024

David and Claudia Harding Foundation

Miscellaneous

Languages: Hungarian (native), English (fluent), German (elementary), Serbian (elementary)
Citizenship: Hungary, Serbia